# SIMATS ENGINEERING

### SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

### COMPUTER SCIENCE AND ENGINEERING PROGRAMME

### CSA3717- SOFTWARE TESTING

### EXPERIMENT 10

## Comprehensive Test Plan – Library Management System

**Aim:**

To prepare a comprehensive test plan for a Library Management System covering functional requirements, testing strategy, test cases, execution, and defect reporting.

## 1. Introduction

This Test Plan defines the testing strategy for the Library Management System. It covers test scope, objectives, design techniques, execution strategy, defect management, risks, and test completion criteria.

## 2. Test Objectives

• Verify functional requirements.

• Detect defects early.

• Validate data integrity and error handling.

• Verify complete end-to-end workflows.

• Ensure acceptable reliability and user experience.

## 3. Test Scope

| In Scope | Out of Scope |
|---|---|
| Functional testing | Third-party internal source-code testing |
| Positive and negative testing | Unrelated external systems |
| Boundary and validation testing | Production monitoring after release |
| Integration and system testing | Future features not in approved requirements |
| Regression testing | Unapproved requirement changes |

## 4. Test Design Techniques

| Technique | Application |
|---|---|
| Equivalence Partitioning | Divide input data into valid and invalid classes. |
| Boundary Value Analysis | Test minimum, maximum and just-outside limits. |
| Decision Table Testing | Verify combinations of conditions and actions. |
| State Transition Testing | Verify states such as Available, Issued, Returned and Reserved. |
| Use Case Testing | Validate complete user workflows. |

## 5. Test Environment

| Item | Specification |
|---|---|
| Application | Library Management System |
| Operating System | Windows 10/11 |
| Browser | Google Chrome / Microsoft Edge |

| | |
|---|---|
| Database | Controlled test database |
| Network | Stable test network |
| Test Data | Valid, invalid, boundary, duplicate and missing data |

## 6. Test Execution Strategy

1. Review requirements and prepare test data.

2. Execute smoke testing on every test build.

3. Execute positive and negative functional tests.

4. Execute integration and system tests.

5. Log defects with reproducible evidence.

6. Retest fixed defects and perform regression testing.

7. Prepare final test summary and obtain sign-off.

## 7. Test Coverage Areas

| **Area** | **Coverage** |
|---|---|
| Book search | Search books by title, author and category. |
| Member management | Create and update valid member records. |
| Book issue | Issue available books to eligible members. |
| Book return | Return books and update availability. |
| Fine calculation | Verify overdue fine rules. |
| Reports | Verify accurate inventory and transaction reports. |

## 8. Entry Criteria

Approved requirements are available; test build is deployed; test environment is ready; test data is prepared; major dependencies are available.

## 9. Exit / Completion Criteria

All planned critical tests are executed; no open Critical defects remain; High-severity defects are resolved or formally accepted; failed tests are analyzed; regression testing is completed; final test summary is prepared.

## 10. Defect Management

Defects are recorded with defect ID, title, linked test case, severity, priority, environment, preconditions, steps to reproduce, expected result, actual result, impact, suggested fix, and status. Life cycle: New → Assigned → In Progress → Fixed → Retesting → Closed/Reopened.

## 11. Risks and Mitigation

| **Risk** | **Mitigation** |
|---|---|
| Test environment unavailable | Maintain a backup environment. |
| Incomplete test data | Prepare controlled test data before execution. |
| Late requirement changes | Perform impact analysis and update test cases. |
| Critical defect near release | Prioritize correction and perform focused regression testing. |

## 12. Conclusion

This Test Plan provides a structured method to evaluate the Library Management System and ensure that the application satisfies its approved requirements with acceptable quality and reliability.